System Capacity & Care Load Analytics for Unaccompanied Children

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Dataset: U.S. Department of Health & Human Services — UAC Program (2023–2025)

Abstract

This paper presents a data-driven analytical framework for monitoring the Unaccompanied Alien Children (UAC) care pipeline managed by the U.S. Department of Health & Human Services. Using daily operational data spanning January 2023 to December 2025 (720 reporting days), we quantify care system load, identify periods of capacity strain, and evaluate the balance between intake and discharge flows.

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1. Introduction

The UAC Program is a federally mandated initiative in which children apprehended at the U.S. border are transferred from CBP custody to HHS care facilities for medical screening, shelter, and sponsor placement. Effective management requires continuous situational awareness of system load, intake pressure, and discharge capacity.

This study addresses the absence of a centralized analytical framework by building a structured analytics pipeline and interactive dashboard using Python, Pandas, Plotly, and Streamlit.

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2. Dataset Description

Attribute	Detail
Source	U.S. Department of Health & Human Services
Date range	January 12, 2023 – December 21, 2025
Raw rows	1,170 (450 empty, 720 valid)
Columns	6 (Date + 5 operational metrics)
Granularity	Daily reporting

2.1 Columns

- **Date** — Reporting date
- **CBP Apprehended** — Daily new intake into CBP custody
- **CBP In Custody** — Active CBP care load
- **CBP Transferred** — Daily flow into HHS system
- **HHS In Care** — Active HHS care load
- **HHS Discharged** — Daily sponsor placements

3. Methodology

3.1 Data Cleaning

- Removed 450 completely empty rows
- Converted date strings to datetime format
- Removed comma formatting from numeric fields
- Sorted chronologically (oldest to newest)
- Validated zero missing values post-cleaning

3.2 Feature Engineering

Six derived metrics were computed:

1. **Total System Load** = CBP Custody + HHS Care
2. **Net Daily Intake** = CBP Transferred − HHS Discharged
3. **Care Load Growth Rate** = Day-over-day % change in HHS care
4. **7-Day Rolling Average** = Smoothed HHS care trend
5. **7-Day Rolling Net Intake** = Smoothed pressure indicator
6. **Cumulative Backlog** = Running total of net daily intake

3.3 Trend Analysis

Data was resampled to weekly and monthly frequencies using Pandas resample(). Stress periods were identified as days exceeding the 75th percentile of HHS care load (threshold: 8,010).

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4. Key Findings

4.1 Crisis Peak — December 2023

The system reached its maximum care load of **11,516 children** on December 20, 2023. The entire top-5 highest care load days occurred in December 2023, indicating a sustained crisis rather than a single-day anomaly. Total system load including CBP custody peaked at 11,762 on the same date.

4.2 Intake Surge — February 2024

Despite declining HHS care load from January 2024, CBP apprehensions peaked in February 2024 with 333 children apprehended in a single day (February 15, 2024). This indicates the pipeline was still receiving high volumes even as HHS worked to discharge the existing population.

4.3 System Collapse — Early 2025

HHS care load underwent a dramatic contraction beginning January 2025:

- January 2025 avg: 4,991
- February 2025 avg: 2,782
- March 2025 avg: 2,165

This 78.4% reduction from peak represents the fastest sustained decline in the dataset, driven primarily by reduced CBP transfers rather than increased discharges.

4.4 Balance Analysis

Year	Transfers In	Discharges	Net Pressure
2023	36,124	66,244	−30,120
2024	52,552	51,689	+863
2025	3,965	6,920	−2,955

The Discharge Offset Ratio of **1.35** across the full timeline indicates the system discharged 35% more children than it received — a net positive humanitarian outcome.

4.5 Stress Period Analysis

- 180 high-stress days (25% of all reporting days)
- All concentrated in the period August 2023 – January 2024
- Stress threshold: 8,010 children (75th percentile)

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5. Dashboard

An interactive Streamlit dashboard was developed with three analytical modules:

1. **System Overview** — KPI cards + care load timeline
2. **CBP vs HHS Comparison** — Dual-axis pipeline analysis
3. **Intake & Backlog Trends** — Pressure and relief visualization

Key dashboard features:

- Date range filtering
- Time granularity toggle (Daily / Weekly / Monthly)
- 7-day rolling average overlay
- Expandable raw data tables

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6. Recommendations

1. **Early warning threshold:** Flag when HHS care load exceeds 8,000 children for proactive resource allocation
2. **Discharge acceleration:** During high-intake months, prioritize sponsor matching to prevent backlog accumulation
3. **Seasonal planning:** August–December shows historically elevated intake — staff and shelter capacity should be pre-positioned
4. **Pipeline monitoring:** CBP-to-HHS transfer lag should be monitored — extended CBP custody indicates HHS capacity constraints

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7. Conclusion

This analytical framework demonstrates that structured data analytics can transform raw operational counts into actionable capacity insights. The UAC care system experienced a full crisis arc from 2023 to 2025 — from peak overload to rapid contraction. Continuous monitoring using the developed dashboard enables data-driven, proactive humanitarian response.

References

- U.S. Department of Health & Human Services, Office of Refugee Resettlement — UAC Program Data
- Python Software Foundation — Pandas, Plotly, Streamlit documentation